

sors and AI will become so fast that humans will undergo a radical evolution.

AI is part of the new frontier in the medical and healthcare industry. But with the proliferation of AI and its advantages in healthcare space, there are speculations that AI will someday replace human doctors in many areas. There are many proofs to support this statement, some of which are covered in this article.

AI winter or AI spring

AI is different from previous technological innovations because it can learn and is more intelligent. Earlier, there were setbacks and frustrations while pursuing AI programs—the term is called AI winter and it first appeared in 1984. But with recent progress in speech recognition, image processing and machine learning, AI has become an essential component of almost every field.

Modern technology has demonstrated many wonders in such areas as self-driving cars, autonomous drones, medicine and healthcare services. Use of AI and the Internet of Medical Things (IoMT) in healthcare is already helping many people.

Now, AI and machine learning are becoming hot topics. Andrew Ng, former director of Google's first AI group, Google Brain and former chief scientist of Baidu, believes that we have entered an eternal AI spring. This is because AI and machine learning have a wide range of applications and a clear roadmap for changing almost all industries. With AI, there are many wonders and surprises in store for us in the coming years. AI is limited only by our imagination!

Medical AI is so hot!

Tens of millions of people die each year from diseases that are treatable. Cancer treatment is not affordable by many. Now, AI is being employed in diagnosing cancer, tuberculosis, skin, eyes, stroke and other conditions, and it is more precise, accurate, faster and



Fig. 1: Vscan portable ultrasound device (Credit: www.gehealthcare.com)

cheaper. AI is not only an industry-disrupting force but an important component that leads to affordability and accessibility for common people.

The current stethoscope could be a dumb thing in the future. For example, an Australian company has developed an AI-enabled electronic stethoscope called Stetsee, which is far better than the current stethoscope. Stetsee can be used as a regular stethoscope, in addition to being used to amplify, filter and record sounds, and to analyse sounds using an accompanying smartphone app.

An ultrasound machine is quite costly, but as technology progresses, its cost will come down and it will be affordable by individuals. There are promising signs of AI solutions in ultrasound imaging. For example, iQ probe with semiconductor chips, developed by Butterfly Network, can be used by patients to perform their own ultrasound tests. Understanding the results of an ultrasound can be hard for untrained patients, but iQ has an AI component in it. As AI gets smarter, risks-associated image interpretation will eventually reduce.

Portable ultrasound that can be accessed on the cloud and smartphones are now available in the market. Israel-based DiA Imaging Analysis, in partnership with GE Healthcare, has developed Vscan, a smartphone-sized ultrasound device. It also collaborates with Google Cloud to offer AI-powered automated ultrasound analysis for patient care, enabling quick analysis and easy accessibility anywhere, anytime.

There is concrete proof that AI is way better than a radiologist in detecting various parameters of the human

body. A recent report showed that Google AI machine can tell vital body details including gender, heart conditions or other cardiovascular problems just by scanning the eyes.

In China, AI is being used to detect diseases that cause blindness. Researchers are using AI to support instant diagnosis

of blindness and diabetes-related eye diseases in the earliest stages. Reports suggest that eye diseases can be treated faster with AI.

AI has had a strong impact on healthcare advancements due to its ability to analyse massive amounts of genomic data, leading to more accurate prevention and treatment of medical conditions on a personalised level. Use of AI robotic surgery has spread rapidly in recent years and, for certain types of operations, it has become the standard.

Imagine that a hospital full of AI-humanoid robots! A robotic doctor pulls your arms and legs to check your body condition; you may feel awkward or uncomfortable at the first instance. But that could be the scenario in the future.

AI and 3D bioprinting

3D bioprinting is the utilisation of 3D printing to print tissues and organs to help research drugs and pills. Researchers at a hospital in Boston have pioneered a 3D bioprinting technique that can create complex structures to replace damaged body tissues. This advanced technique could lead to a huge demand for bioprinting complex artificial tissues.

AI-powered medical imaging software developed by Aether has given a significant boost to 3D organ printing.

The thyroid gland printer from Russia-based 3D Bioprinting Solutions is shown in Fig. 2.

AI and DNA sequencing

Genome editing (or gene editing) is a type of genetic engineering. The worlds of genetics and AI are set to explode soon. With genetic engineer-